

Research Notes on Adam optimizer

Adam optimizer

$$m_t = \beta_1 m_{t-1} + (1 - \beta_1) g_t$$

$$\theta_t = \theta_{t-1} - \frac{\eta \hat{m}_t}{\sqrt{\hat{v}_t + \epsilon}}$$

>> Section 1

where the parameter w that minimizes $Q(w)$ is to be estimated. Each summand function Q_i is typically associated with the i -th observation in the data set (used for training). In classical statistics, sum-minimization problems arise in least squares and in maximum-likelihood estimation (for independent observations). The general class of estimators that arise as minimizers of sums are called M-estimators. However, in statistics, it has been long recognized that requiring even local minimization is too restrictive for some problems of maximum-likelihood estimation. Therefore, contemporary statistical theorists often consider stationary points of the likelihood function (or zeros of its derivative, the score function, and other estimating equations). The sum-minimization problem also arises for empirical risk minimization. There, $Q_i(w)$ is the value of the loss function at i -th example, and $Q(w)$ is the empirical risk. When used to minimize the above function, a standard (or "batch") gradient descent method would perform the following iterations:

>> Section 2

where $x_j'w = x_{j,1}w_1 + x_{j,2}w_2 + \dots + x_{j,p}w_p$ indicates the inner product. Note that x could have "1" as the first element to include an intercept. Classical stochastic gradient descent proceeds as follows:

>> Section 3

for $\Sigma(w) = \frac{1}{n^2} (\sum_{i=1}^n Q_i(w) - Q(w)) (\sum_{i=1}^n Q_i(w) - Q(w))^T$ where $\int_0^t dB_s$ denotes the Ito-integral with respect to a Brownian motion is a more precise approximation in the sense that there exists a constant $C > 0$ such that

>> Section 4

Further reading Bottou, Léon (2004), "Stochastic Learning", Advanced Lectures on Machine Learning, LNAI,

vol. 3176, Springer, pp. 146–168, ISBN 978-3-540-23122-6 Buduma, Nikhil; Locascio, Nicholas (2017), "Beyond Gradient Descent", Fundamentals of Deep Learning : Designing Next-Generation Machine Intelligence Algorithms, O'Reilly, ISBN 9781491925584 LeCun, Yann A.; Bottou, Léon; Orr, Genevieve B.; Müller, Klaus-Robert (2012), "Efficient BackProp", Neural Networks: Tricks of the Trade, Springer, pp. 9–48, ISBN 978-3-642-35288-1 Spall, James C. (2003), Introduction to Stochastic Search and Optimization, Wiley, ISBN 978-0-471-33052-3

>> Section 5

$$v(w, t) := \gamma v(w, t-1) + (1 - \gamma) (\nabla Q_i(w))^2$$

>> Section 6

Approximations in continuous time For small learning rate η stochastic gradient descent $(w_n)_{n \in \mathbb{N}_0}$ can be viewed as a discretization of the gradient flow ODE

>> Section 7

Sign-based stochastic gradient descent Even though sign-based optimization goes back to the aforementioned Rprop, in 2018 researchers tried to simplify Adam by removing the magnitude of the stochastic gradient from being taken into account and only considering its sign. This results in a significantly lower communication cost of transferring gradients from workers to the parameter server. In this sense, it serves to better compress the gradient information, while having comparable convergence to standard SGD.

>> Section 8

$$\max_{k=0, \dots, \lfloor T/\eta \rfloor} |E[g(w_k)] - E[g(W_{k\eta})]| \leq C \eta^2 \cdot \max_{k=0, \dots, \lfloor T/\eta \rfloor} |E[g(w_k)] - E[g(W_{k\eta})]| \leq C \eta^2.$$

>> Section 9

Notable applications Stochastic gradient descent is a popular algorithm for training a wide range of models in machine learning, including (linear) support vector machines, logistic regression (see, e.g., Wovpal Wabbit) and graphical models. When combined with the backpropagation algorithm, it is the de facto standard algorithm for training artificial neural networks. Its use has been also reported in the Geophysics community, specifically to applications of Full Waveform Inversion (FWI). Stochastic gradient descent competes with the L-BFGS algorithm, which is also widely used. Stochastic

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gradient descent has been used since at least 1960 for training linear regression models, originally under the name ADALINE. Another stochastic gradient descent algorithm is the least mean squares (LMS) adaptive filter.

>> Section 10

$$w^{\text{new}} := \arg \min_w \{ Q_i(w) + \frac{1}{2} \eta \|w - w^{\text{old}}\|^2 \}.$$

>> Section 11

This equation is implicit since w^{new} appears on both sides of the equation. It is a stochastic form of the proximal gradient method since the update can also be written as:

>> Section 12

$$w(t) := w(t-1) - \eta m^w(t) + \varepsilon$$

>> Section 13

Linear regression Suppose we want to fit a straight line $\hat{y} = w_1 + w_2 x$ to a training set with observations $((x_1, y_1), (x_2, y_2), \dots, (x_n, y_n))$ and corresponding estimated responses $(\hat{y}_1, \hat{y}_2, \dots, \hat{y}_n)$ using least squares. The objective function to be minimized is

>> Section 14

$$w := w - \eta \nabla Q(w) = w - \eta \sum_{i=1}^n \nabla Q_i(w).$$

>> Section 15

$$dW_t = -\nabla(Q(W_t) + \frac{1}{4} \eta \|\nabla Q(W_t)\|^2) dt + \eta \Sigma(W_t)^{1/2} dB_t,$$

>> Section 16

This procedure will remain numerically stable virtually for all η as the learning rate is now normalized. Such comparison between classical and implicit stochastic gradient descent in the least squares problem is very similar to the comparison between least mean squares (LMS) and normalized least mean squares filter (NLMS). Even though a closed-form solution for ISGD is only possible in least squares, the procedure can be

efficiently implemented in a wide range of models. Specifically, suppose that $Q_i(w)$ depends on w only through a linear combination with features x_i , so that we can write $\nabla_w Q_i(w) = -q(x_i'w)x_i$, where $q(\cdot) \in \mathbb{R}$ may depend on x_i, y_i as well but not on w except through $x_i'w$. Least squares obeys this rule, and so does logistic regression, and most generalized linear models. For instance, in least squares, $q(x_i'w) = y_i - x_i'w$, and in logistic regression $q(x_i'w) = y_i - S(x_i'w)$, where $S(u) = e^u / (1 + e^u)$ is the logistic function. In Poisson regression, $q(x_i'w) = y_i - e^{x_i'w}$, and so on. In such settings, ISGD is simply implemented as follows. Let $f(\xi) = \eta q(x_i'w^{\text{old}} + \xi x_i)$, where ξ is scalar. Then, ISGD is equivalent to: